#4 · 10.5% of marks

Ch 8 — Mutual Funds

Tested every attempt: NAV working, holding-period return (payout vs reinvest), dividend vs bonus plans, back-solving issue price, closed-end premium / discount.

1 Set-up & structure

- → *MF = **trust** pooling savings via **units**; invests in capital & money market; income & gains shared in proportion to units*
- → *Four constituents: **Sponsor** → **Trust (trustees)** → **AMC** → **Custodian**, all SEBI-registered. Trust deed registered under the Registration Act, 1908*
- → *Sponsor: **5 years** in financial services, holds **≥ 40%** of AMC net worth. AMC net worth **₹100 crore** (SM). Trustees ⅔ **independent**; AMC board **50% independent**. AMC removed by majority of trustees or **75%** of unit holders*
- → *Custodian: SEBI-registered, **independent of sponsor & AMC** (DP for demat)*

2 Classification

Basis	Types
Structure	***Open-ended** (buy / redeem any time with fund; no maturity) · **Close-ended** (NFO then exchange; exit on exchange / maturity / repurchase window) · **Interval** (need not list; in-built redemption window; can issue fresh units at NAV in the interval; no fixed maturity)*
Asset	*Equity (**growth** — long-term appreciation; **aggressive** — IPOs, speculative; **income** — high-dividend stocks) · Debt / bond (risks: **I-C-P-R** — interest rate, credit, prepayment, reinvestment) · Money market / liquid · Gilt · Hybrid / balanced*
Special	***Index** · **International** (raises in India, invests abroad) vs **Offshore** (outside India, raises globally to invest in India) · **Capital protection oriented** (protection from portfolio structure, not guarantee; close-ended, listed, structure rated) · **Quant** (rule-based models; no bias, low cost; relies on history) · **Fund of funds** · **ETF** · **Arbitrage** (spot vs futures) · **Sector** vs **thematic***
Ownership	***Public sector** (sponsored by a public-sector company / bank) · **Private sector** · **Foreign** (foreign company operating from India, raises and invests in India)*
Diversified equity (SM 3.2)	***Flexicap** · **Multicap** · **Contra** (out-of-favour / undervalued companies; blends growth + value; needs aggressive risk appetite) · **Index** · **Dividend yield** (yield above the index's; less volatile than growth stocks)*
Others (SM 3)	***Thematic** — broader than sector (multi-sector, international, commodity); market may take long to recognise the theme · **Arbitrage** — open-ended, low-volatility, cash equities + derivatives + debt; spot vs futures differential · **Cash fund** — open-ended liquid, very short-dated debt & money market, growth & dividend options*
Schemes	***Flexicap** (any proportion) vs **Multicap** (≥ 25% each large / mid / small) · **FMP** (no interest-rate risk, still credit risk; close-ended, illiquid) · **ELSS** (80C, 3-yr lock-in; SIP preferred)*
Options	***Growth** (nothing paid) · **Dividend payout** (NAV falls by dividend) · **Dividend reinvestment** (extra units at ex-dividend NAV)*
Plans	***Direct plan** (from 1 Jan 2013) — no distribution expense / trail → lower TER, **higher NAV** than regular*

⚠ TRAP

*Interval ≠ close-ended. CPOF ≠ capital **guaranteed**. FMP is free of **interest-rate** risk, **not** credit risk.*

3 ETF vs index fund; hedge fund vs MF

ETF	Listed; trades all day at real-time prices close to NAV (open-ended fund deals once a day at NAV); **authorised participants** act as market makers; low cost, tax-efficient, no paperwork. Types: index, commodity (gold), bond, currency
Hedge fund vs MF	Lightly regulated private vehicle for selected clients; little public disclosure; **performance fee** (losses borne by investors); uses **leverage, long / short, derivatives**

High-water mark fee (SM: ANP Plan): ₹20 cr; 0.10% fixed + 2% above ₹21 cr. Return 29% → value 25.8 cr → fee ₹2 L + 2% × 4.8 cr = **₹11.60 L**; return 4.5% → 20.9 cr < 21 → **₹2 L only**. Trap: fee on excess over HWM, not on the year's gain.

4 Advantages (12) & drawbacks (4)

Advantages

Professional management · diversification · convenient administration · higher returns · **low cost** (max **2.50%** for equity funds; excess borne by AMC) · liquidity · transparency · other benefits (SIP / SWP / switch) · highly regulated · economies of scale · flexibility · convenience

Drawbacks

1. No guarantee of return (under-performance, entry / exit timing, crashes) · 2. Diversification caps upside · 3. Picking the right fund is hard; past performance no guide · 4. Costs; AMC fees not linked to performance

5 Key terms

Term	Meaning / formula
NAV	Net assets ÷ units outstanding. Net assets = MV of investments + receivables + accrued income + other assets − accrued expenses − payables − other liabilities
Loads	Public offer price = NAV ÷ (1 − entry load) · Redemption = NAV ÷ (1 + exit load) [or NAV × (1 − exit load) where the question says]. Exit load goes back to the **scheme**. No entry load now
Required MF return	$r_2 = r_1 \div (1 - \text{initial expenses}) + \text{recurring expenses}$ (10%, 6%, 2% → **12.64%**)
Expense ratio (TER)	Mgmt & advisory fees, travel, audit, custodian, registrar — **excludes brokerage**; charged daily (1/365) before NAV
Trail commission	0.30–0.75% p.a. on daily average AUM, paid monthly, while money stays invested
Side pocketing	Stressed / downgraded assets moved to a segregated portfolio; main NAV reflects liquid assets; protects continuing holders (IL&FS). Recovery paid only to holders on segregation date
Tracking error	$\sqrt{[\Sigma(d - \bar{d})^2 \div (n - 1)]}$, d = fund − benchmark return. Causes: transaction cost, AMC fees, expenses, cash holdings, sampling. Higher TE = higher risk even if outperforming
Dividend equalisation	New investor pays / exiting investor gets undistributed income per unit: issue = NAV + entry load + income p.u.; repurchase = NAV − exit load + income p.u.

6 Return formulas

HPR = $(NAV_1 - NAV_0 + \text{income distributed} + \text{capital-gain distributed}) \div NAV_0$

Reinvestment: extra units = distribution ÷ reinvestment NAV; return = (total units × closing NAV − cost) ÷ cost

Effective yield p.a. = [(units × closing NAV − investment) + dividends] ÷ investment × **365 ÷ days**

Annualise monthly return × 12 (simple) unless told to compound; **real return** = (1 + nominal) ÷ (1 + inflation) − 1

Pattern	Method
Payout vs reinvest (Sep 2025 Q3(b))	Compute both; reinvest at the given (average) NAV; pick higher return
Entry load + appreciation + distribution reinvested + exit load + tax + inflation (Sep 2025 Q5(b))	Units = amount ÷ POP; end value = units × NAV × (1 − exit load); tax on gain; real return
Dividend % declared	Dividend = units × **face value ₹10** × % (not × NAV)
Find month-end NAV from annual return (PYQ May 2024)	Monthly = 16% ÷ 12 = 1.33%; 0.0133 = (NAV − 78.50 + 0.70) ÷ 78.50 → **₹78.84**
Days invested from yields	Days = period yield × 365 ÷ yield p.a.; count back

7 Dividend vs bonus vs growth plans

→ Build a **units table** per plan to redemption; value = units × closing NAV; deduct STT / tax on short-term units only; annual return = gain ÷ investment × 12 ÷ months

→ **Back-solve issue price** (Jan 2026 Q1(a)): value now = investment × (1 + yield × years); units now = value ÷ NAV; **undo bonuses latest first** (× b ÷ (a + b)); issue price = investment ÷ original units. SM: ₹16 L, 17.5% × 5 → ₹30 L ÷ ₹100 = 30,000 → 24,000 → 20,000 → **₹80**

8 NAV statement (new fund, one month)

1. Cash = amount raised − underwriting commission − marketing / issue costs − investments bought + sale proceeds + dividends received − expenses **paid**
2. Net assets = cash + market value of holdings + receivables (uncollected dividend) − **arrears / outstanding expenses**
3. Distribution = % of **realised** earnings only (sale gains + dividends), deducted from cash
4. NAV per unit = net assets ÷ units (underwritten issue → all units outstanding)

⚠ TRAP

Only expenses **paid** leave cash; arrears are deducted from net assets. Units issued on a cheque = cheque ÷ **unrounded** current NAV; NAV unchanged that day.

HIGH

9 Evaluating a mutual fund

Quantitative (5)

1. Risk-adjusted returns · 2. Benchmark returns (alpha) · 3. Comparison with peers · 4. Returns across economic & market cycles · 5. Financial measures — expense ratio, Sharpe, Treynor, Sortino

Qualitative (3)

1. Quality of portfolio (blue chips, diversification, style; debt: credit quality, maturity, modified duration; low turnover) · 2. Track record & competence of fund manager · 3. Credibility of fund house team

Measure	Formula / note
Sharpe	$(R_p - R_f) \div \sigma_p$ — total risk
Treynor	$(R_p - R_f) \div \beta$ — systematic risk
Sortino	SM: variation of Sharpe using **downside deviation** instead of total σ in the denominator
Jensen α	$R_p - [R_f + \beta(R_m - R_f)]$
M^2 (not in SM)	$R_f + \text{Sharpe}_p \times \sigma_m$ — compare with R_m
Information ratio (not in SM)	$(R_p - R_{benchmark}) \div$ tracking error

10 Reverse-engineering β and NAV

Given	Get
Sharpe & σ → excess return; with Treynor	**β = (Sharpe × σ) ÷ Treynor**
ρ, σ_p, market variance	$\beta = \rho\sigma_p \div \sigma_m$ ("variance 9%" → $\sigma_m = 3$)
Sharpe, Treynor, σ_m	σ = excess ÷ Sharpe; β = excess ÷ Treynor; $\rho = \beta\sigma_m \div \sigma$; **$R^2 = \rho^2$**
Jensen α & Treynor	CAPM return = actual − α; solve R_f and β simultaneously

Expected NAV (SM): NAV ₹70.71, 99% equity, Sharpe 2, σ 11.25, Treynor 15 → β = 22.5 ÷ 15 = 1.5. Market −5% → equity ₹70 × (1 − 7.5%) = 64.75; cash 0.71 − monthly cost 0.25 = 0.46 → **NAV ₹65.21**. Trap: **don't apply β to the cash part**.

11 Closed-ended fund premium / discount

Market price = NAV × (1 ± premium / discount). Investor's return uses **market prices**, so it beats NAV growth if the discount narrows. PYQ May 2026: geometric mean return 3.771% vs NAV growth 2.76%. Geometric mean = $[(1 + r_1)(1 + r_2)\ldots]^{1/n} - 1$.

12 Role of fund manager & FIIs

- → **Active** fund: outperform, generate positive alpha via research & stock picking. **Passive**: match index with minimum tracking error
- → Other roles (**C-M-W-O**): **C**ompliances (SEBI / AMFI, reporting), constant **M**onitoring of performance, **W**ealth creation & protection, control over **O**utsourced work
- → **FIIs** buy units directly from the issuer or through a registered broker on a recognised exchange within SEBI limits; benefits: **better corporate governance**, **more competition**, **capital inflow** (jobs & forex)
- → Fund hedge with index futures: contracts = β × equity value ÷ (futures × lot) — **cash not hedged** (₹90 cr, β 1.1, 4,300 × 50 → 4,605)

13 Formula plans — constant ratio plan

- → Total value = bonds + units × NAV; when NAV move hits the trigger, **reset to the fixed ratio** — buy units after a fall, sell after a rise
- → ICAI checks each NAV against the **previous** one (32 → 38 = +18.75% triggers)
- → Wins in **choppy** markets ending near the start (sells high, buys low); loses in a strong one-way rise (keeps selling the winner)

14 SIP / STP / SWP & vocabulary

- → SIP: fixed amount in; average cost = total ₹ ÷ total units = **harmonic mean** of NAVs ≤ average NAV (₹10,000 at 20, 16, 25, 20 → ₹19.75 vs ₹20.25)
- → SWP: fixed amount out · STP: fixed amount moved scheme to scheme
- → NFO · AUM · folio · **cut-off time** (before → that day's NAV; after → next day's)
- → Gilt fund: rates expected to fall → **lengthen duration**; to rise → shorten

15 Correlation — how Ch 8 actually connects

Ch 8 idea	The actual link (app-written cross-reference; the Ch 8 side is SM)
Sharpe / Treynor / Jensen	**Ch 6**: same formulas; Sharpe for an undiversified investor (σ), Treynor for a diversified one (β), Jensen α positive = fund beat CAPM.
Debt / gilt fund NAV	**Ch 5**: rates ↑ → bond prices ↓ → NAV ↓; % change ≈ −modified duration × Δy, so longer-duration funds swing more.
Hedging a fund	**Ch 9**: sell index futures = Value × β ÷ (futures price × lot) to protect NAV against a market fall.
Index funds / ETFs	**Ch 6** passive strategy and **Ch 4** efficient-market logic — low cost, tracks index.
Units held by a company	**FR Ind AS 109**: MF units generally fail SPPI and are not equity instruments for the OCI election → FVTPL.

> ✍ **Write in exam**
> "NAV per unit = (market value of investments + receivables + accrued income + other assets − accrued expenses − payables − other liabilities) ÷ number of units outstanding."

Past papers

⑧ Mutual Funds — what ICAI asked (Sep 2025, Jan 2026, May 2026)

37 marks *set across the last three attempts (10.5% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.*

Attempt	Question	Marks	Concept tested
Sep 2025	MCQ CS-II 4-6	6	Mutual fund accounts: closing net cash balance, total NAV, and NAV per unit after a dividend per unit is distributed (MCQ)
Sep 2025	Q3(b)	4	Mutual fund holding period return: payout of dividend and capital gain versus reinvestment at average NAV; choose the better option (Numerical)
Sep 2025	Q5(b)	7	Mutual fund: units bought after entry load, NAV after appreciation, distribution reinvested at year-end NAV, return and real return after exit load, tax and inflation (Numerical)
Jan 2026	Q1(a)	6	Dividend (bonus) scheme: back-solve opening NAV per unit from average annual yield, two bonus issues (1:4 and 2:5) and closing NAV (Numerical)
Jan 2026	Q6(a)	6	New fund offer: closing NAV per unit after underwriting commission, marketing cost, gains, regulated expenses, income distribution policy (Numerical)
May 2026	Q3(b)	4	Dividend plan versus bonus plan of a mutual fund: total return and holding period return after entry/exit loads; which plan is better (Numerical)
May 2026	Q4(b)	4	Closed-ended fund: market price from NAV and premium/discount, total, average and geometric return per period for the investor, geometric growth of NAV, interpretation (Numerical)

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Ⓣ Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. Explain how to establish a Mutual Fund.

ICAI answer

1. Establishment of a Mutual Fund: A mutual fund is required to be registered with the Securities and Exchange Board of India (SEBI) before it can collect funds from the public. All mutual funds are governed by the same set of regulations and are subject to monitoring and inspections by the SEBI. The Mutual Fund has to be established through the medium of a sponsor. A sponsor means any body corporate which, acting alone or in combination with another body corporate, establishes a mutual fund after completing the formalities prescribed in the SEBI's Mutual Fund Regulations.

The role of sponsor is akin to that of a promoter of a company, who provides the initial capital and appoints the trustees. The sponsor should be a body corporate in the business of financial services for a period not less than 5 years, be financially sound and be a fit party to act as sponsor in the eyes of SEBI.

The Mutual Fund has to be established as either a trustee company or a Trust, under the Indian Trust Act and the instrument of trust shall be in the form of a deed. The deed shall be executed by the sponsor in favour of the trustees named in the instrument of trust. The trust deed shall be duly registered under the provisions of the Indian Registration Act, 1908. The trust deed shall contain clauses specified in the Third Schedule of the Regulations.

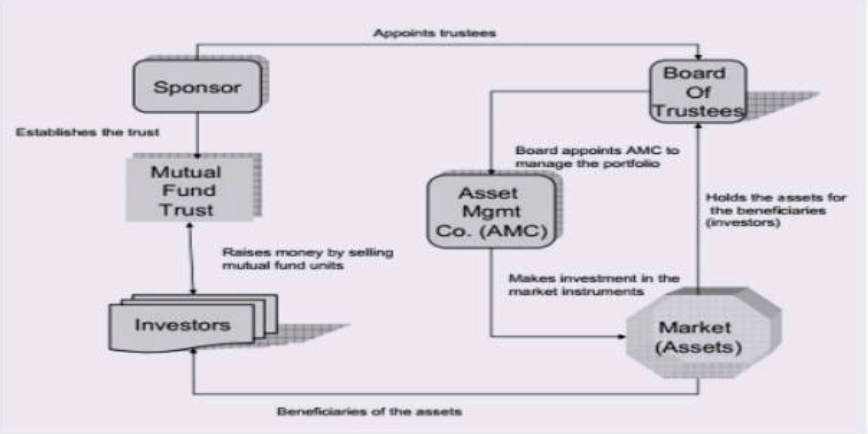
An Asset Management Company, who holds an approval from SEBI, is to be appointed to manage the affairs of the Mutual Fund and it should operate the schemes of such fund. The Asset Management Company is set up as a limited liability company, with a minimum net worth of 100 crores.

The sponsor should contribute at least 40% to the net worth of the Asset Management Company. The Trustee should hold the property of the Mutual Fund in trust for the benefit of the unit holders.

SEBI regulations require that at least two-thirds of the directors of the Trustee Company or board of trustees must be independent, that is, they should not be associated with the sponsors. Also, 50 per cent of the directors of AMC must be independent. The appointment of the AMC can be terminated by majority of the trustees or by 75% of the unit holders of the concerned scheme.

The AMC may charge the mutual fund with Investment Management and Advisory fees subject to prescribed ceiling. Additionally, the AMC may get the expenses on operation of the mutual fund reimbursed from the concerned scheme.

The Mutual fund also appoints a custodian, holding valid certificate of registration issued by SEBI, to have custody of securities held by the mutual fund under different schemes. In case of dematerialized securities, this is done by Depository Participant. The custodian must be independent of the sponsor and the AMC.



Theory Q2

2. What are the advantages of investing in Mutual Funds?

ICAI answer: "Please refer paragraph 4" — SM paragraph 4 reproduced

4. ADVANTAGES OF MUTUAL FUND

(a) **Professional Management:** The funds are managed by skilled and professionally experienced managers backed by a team of Research Analysts.

(b) **Diversification:** Mutual Funds invest into many securities and offer diversification which reduces the concentration risk.

(c) **Convenient Administration:** There are no administrative risks of share transfer, as many of the Mutual Funds offer services in a demat form which saves investor's time and prevents delay.

(d) **Higher Returns:** Over a medium to long-term investment horizon, investors get higher returns in Mutual Funds as compared to other avenues of investment. However, investors are cautioned that such very high returns during the exceptional bull phase of the market like IT

boom or Infrastructure boom should not be considered as regular returns and therefore one should look at the average returns provided by the Mutual Funds particularly in the equity schemes over a long period of time.

(e) **Low Cost of Management:** SEBI has prescribed maximum limit of charging 2.50% for Equity Mutual Funds. No Mutual Fund can increase the cost beyond prescribed limits of 2.5% maximum and any extra cost of management is to be borne by the AMC.

(f) **Liquidity:** In all the open ended funds, liquidity is provided by direct sales / repurchase by the Mutual Fund and in case of close ended funds, the liquidity is provided by listing the units on the Stock Exchange.

(g) **Transparency:** The SEBI Regulations now compel all the Mutual Funds to disclose their portfolios on a half-yearly basis. However, many Mutual Funds disclose their Scheme Portfolio on a quarterly or monthly basis to their investors. The NAVs are calculated on a daily basis in case of open ended funds and are published through AMFI in the newspapers.

(h) **Other Benefits:** Mutual Funds provide systematic withdrawal and systematic investment plans according to the need of the investors. The investors can also switch from one scheme to another without any restrictions except in case of Tax Savings Fund which restricts switch out for first 3 years of its investments.

(i) **Highly Regulated:** Mutual Funds all over the world are highly regulated and in India all Mutual Funds are registered with SEBI and are strictly regulated as per the Mutual Fund Regulations which provide high level of investor protection.

(j) **Economies of scale:** The way mutual funds are structured gives it a natural advantage. The "pooled" money from numerous investors ensures that mutual funds enjoy economies of scale; it is cheaper compared to investing directly in the capital markets which involves higher charges. This also allows retail investors access to participation in the Capital Market which otherwise is difficult for them go directly.

(k) **Flexibility:** One of the biggest advantages of a Mutual Fund Scheme is its flexibility. An investor can opt for Systematic Investment Plan (SIP), Systematic Withdrawal Plan etc. to plan his cash flow requirements as per his convenience. The wide range of schemes being launched in India by different mutual funds also provides an added flexibility to the investor to plan his portfolio accordingly.

(l) **Convenience:** It is very convenient & easy to invest & disinvest from Mutual Fund Schemes specially through digital transaction portals.

Theory — TYK with ICAI answers

ICAI answer: "Please refer paragraph 4" — SM paragraph 4 reproduced

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Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical Q1

1. Mr. A can earn a return of 16 per cent by investing in equity shares on his own. Now he is considering a recently announced equity based mutual fund scheme in which initial expenses are 5.5 per cent and annual recurring expenses are 1.5 per cent. How much should the mutual fund earn to provide Mr. A return of 16 per cent?

Solution — ICAI

1. Personal earnings of Mr. A = R_1 = 16%

Mutual Fund earnings = R_2

$$R_2 = \frac{1}{1 - \text{Initial expenses (\%)}} R_1 + \text{Recurring expenses (\%)}$$

$$= \frac{1}{1-0.055} \times 16\% + 1.5\%$$

= 18.43%

Mutual Fund earnings = 18.43%

Practical Q2

2. The unit price of Equity Linked Savings Scheme (ELSS) of a mutual fund is ₹ 10/-. The public offer price (POP) of the unit is ₹ 10.204 and the redemption price is ₹ 9.80.

Calculate:

(i) Front-end Load

(ii) Back end Load

Solution — ICAI

2. Public Offer Price = NAV/ (1 – Front end Load)

Public Offer Price: ₹ 10.204 and NAV: ₹ 10

Accordingly,

10.204 = 10/(1 – F)

F = 0.0199 say 2%

Redemption Price = NAV/ (1 + Back End Load)

₹ 9.80 = 10/ (1 + Back End Load)

B = 0.0204 i.e. 2.04%

Alternative

(i) Front End Load = $\frac{10.204 - 10.00}{10.00}$ = 0.0204 or 2.04%

(ii) Exit Load = $\frac{10.00 - 9.80}{10.00}$ = 0.020 or 2.00%

Practical Q3

3. A mutual fund that had a net asset value of ₹ 20 at the beginning of month - made income and capital gain distribution of ₹ 0.0375 and ₹ 0.03 per share respectively during the month, and then ended the month with a net asset value of ₹ 20.06. Calculate monthly return.

Solution — ICAI

3. Calculation of Monthly Return on the Mutual Funds

$$r = \left[\frac{(NAV_t - NAV_{t-1}) + I_t + G_t}{NAV_{t-1}}\right]$$

Where,

r = Return on the mutual fund

NAV_t = Net assets value at time period t

NAV_{t-1} = Net assets value at time period t – 1

I_t = Income at time period t

G_t = Capital gain distribution at time period t

$$= \left[\frac{(₹\ 20.06 - ₹\ 20.00) + (₹\ 0.0375 + ₹\ 0.03)}{20}\right]$$

$$= \frac{0.06 + 0.0675}{20}$$

$$= \frac{0.1275}{20} = 0.006375$$

Or, r = 0.6375% p.m.

Or = 7.65% p.a.

Practical Q4

4. An investor purchased 300 units of a Mutual Fund at ₹ 12.25 per unit on 31st December, 2009. As on 31st December, 2010 he has received ₹ 1.25 as dividend and ₹ 1.00 as capital gains distribution per unit.

Required:

(i) The return on the investment if the NAV as on 31st December, 2010 is ₹ 13.00.

(ii) The return on the investment as on 31st December, 2010 if all dividends and capital gains distributions are reinvested into additional units of the fund at ₹ 12.50 per unit.

Solution — ICAI

4. Return for the year (all changes on a per year basis)

Particulars	₹ /Unit
Change in price (₹ 13.00 – ₹ 12.25)	0.75
Dividend received	1.25
Capital gain distribution	1.00
Total Return	3.00

Return on investment = $\frac{3.00}{12.25} \times 100 = 24.49\%$

Alternatively, it can also be computed as follows:

$$\frac{(NAV_1 - NAV_0) + D_1 + CG_1}{NAV_0} X 100$$

$$= \frac{(13 - 12.25) + 1.25 + 1.00}{12.25} X 100$$

= 24.49%

If all dividends and capital gain are reinvested into additional units at ₹ 12.50 per unit the position would be.

Total amount reinvested = ₹ 2.25 × 300 = ₹ 675

Additional units added = $\frac{₹\ 675}{12.50}$ = 54 units

Value of 354 units as on 31-12-2010 = ₹ 4,602

Price paid for 300 units on 31-12-2009 (300 × ₹ 12.25) = ₹ 3,675

Return = $\frac{₹\ 4,602 - ₹\ 3,675}{₹\ 3,675} = \frac{₹\ 927}{₹\ 3,675}$ = 25.22%

Practical QS

5. SBI mutual fund has a NAV of ₹ 8.50 at the beginning of the year. At the end of the year NAV increases to ₹ 9.10. Meanwhile fund distributes ₹ 0.90 as dividend and ₹ 0.75 as capital gains.

   (i) What is the fund's return during the year?

   (ii) Had these distributions been re-invested at an average NAV of ₹ 8.75 assuming 200 units were purchased originally. What is the return?

Solution — ICAI

5. (i) Normal Return for the year (all changes on a per year basis)

Particulars	₹ /Unit
Change in price (₹ 9.10 – ₹ 8.50)	0.60
Dividend received	0.90
Capital gain distribution	0.75
Total Return	2.25

Return on investment = $\frac{2.25}{8.50} \times 100 = 26.47\%$

(ii) If all dividends and capital gain are reinvested into additional units at ₹ 8.75 per unit the position would be.

Total amount reinvested = ₹ 1.65 × 200 = ₹ 330

Additional units added = $\frac{₹\,330}{8.75}$ = 37.71 units

Value of 237.71 units at end of year = ₹ 2,163.16

Price paid for 200 units in beginning of the year (200 × ₹ 8.50) = ₹ 1,700

Return = $\frac{₹\ 2,163.16 - ₹\ 1,700}{₹\,1,700} = \frac{₹\,463.16}{₹\,1,700} = 27.24\%$

Practical Q6

6. The following information is extracted from Steady Mutual Fund's Scheme:

- Asset Value at the beginning of the month - ₹ 65.78
- Annualised return -15 %
- Distributions made in the nature of Income & Capital gain (per unit respectively). - ₹ 0.50 and ₹ 0.32

You are required to:

(i) Calculate the month end net asset value of the mutual fund scheme (limit your answers to two decimals).

(ii) Provide a brief comment on the month end NAV.

Solution — ICAI

6. (i) Calculation of NAV at the end of month:

Given Annual Return = 15%

Hence Monthly Return = 1.25% (r)

$$r = \frac{(NAV_t - NAV_{t-1}) + I_t + G_t}{NAV_{t-1}}$$

$$0.0125 = \frac{(NAV_t - ₹\ 65.78) + ₹\ 0.50 + ₹\ 0.32}{₹\ 65.78}$$

$0.82 = NAV_t - ₹\ 64.96$

$NAV_t = ₹\ 65.78$

(ii) There is no change in NAV.

Practical Q7

7. Cinderella Mutual Fund has the following assets in Scheme Rudolf at the close of business on 31st March,2014.

Company	No. of Shares	Market Price Per Share
Nairobi Ltd.	25000	₹ 20
Dakar Ltd.	35000	₹ 300
Senegal Ltd.	29000	₹ 380
Cairo Ltd.	40000	₹ 500

The total number of units of Scheme Rudolf are 10 lacs. The Scheme Rudolf has accrued expenses of ₹ 2,50,000 and other liabilities of ₹ 2,00,000. Calculate the NAV per unit of the Scheme Rudolf.

Solution — ICAI

7.

Shares	No. of shares	Price	Amount (₹)
Nairobi Ltd.	25,000	20.00	5,00,000
Dakar Ltd.	35,000	300.00	1,05,00,000
Senegal Ltd.	29,000	380.00	1,10,20,000
Cairo Ltd.	40,000	500.00	2,00,00,000
			4,20,20,000

Less: Accrued Expenses	2,50,000
Other Liabilities	2,00,000
Total Value	4,15,70,000
No. of Units	10,00,000
NAV per Unit (4,15,70,000/10,00,000)	41.57

Practical Q8

8. A Mutual Fund Co. has the following assets under it on the close of business as on:

Company	No. of Shares	1st February 2012 Market price per share ₹	2nd February 2012 Market price per share ₹
L Ltd	20,000	20.00	20.50
M Ltd	30,000	312.40	360.00
N Ltd	20,000	361.20	383.10
P Ltd	60,000	505.10	503.90

Total No. of Units 6,00,000

(i) Calculate Net Assets Value (NAV) of the Fund.

(ii) Following information is given:

Assuming one Mr. A, submits a cheque of ₹ 30,00,000 to the Mutual Fund and the Fund manager of this company purchases 8,000 shares of M Ltd; and the balance amount is held in Bank. In such a case, what would be the position of the Fund?

(iii) Find new NAV of the Fund as on 2nd February 2012.

Solution — ICAI

8. (i) NAV of the Fund

$$= \frac{₹\ 4,00,000 + ₹\ 93,72,000 + ₹\ 72,24,000 + ₹\ 3,03,06,000}{6,00,000}$$

$$= \frac{₹\ 4,73,02,000}{6,00,000} = ₹\ 78.8366 \text{ rounded to } ₹\ 78.84$$

(ii) The revised position of fund shall be as follows:

Shares	No. of shares	Price	Amount (₹)
L Ltd.	20,000	20.00	4,00,000
M Ltd.	38,000	312.40	1,18,71,200
N Ltd.	20,000	361.20	72,24,000
P Ltd.	60,000	505.10	3,03,06,000
Cash			5,00,800
			5,03,02,000

No. of units of fund = $6,00,000 + \frac{30,00,000}{78.8366} = 6,38,053$

(iii) On 2nd February 2012, the NAV of fund will be as follows:

Shares	No. of shares	Price	Amount (₹)
L Ltd.	20,000	20.50	4,10,000
M Ltd.	38,000	360.00	1,36,80,000
N Ltd.	20,000	383.10	76,62,000
P Ltd.	60,000	503.90	3,02,34,000
Cash			5,00,800
			5,24,86,800

NAV as on 2nd February 2012 = $\frac{₹\ 5,24,86,800}{6,38,053}$ = ₹ 82.26 per unit

Practical Q9

9. On 1st April 2009 Fair Return Mutual Fund has the following assets and prices at 4.00 p.m.

Shares	No. of Shares	Market Price Per Share (₹)
A Ltd.	10000	19.70
B Ltd.	50000	482.60
C Ltd.	10000	264.40
D Ltd.	100000	674.90
E Ltd.	30000	25.90
No. of units of funds		8,00,000

Please calculate:

(a) NAV of the Fund on 1st April 2009.

(b) Assuming that on 1st April 2009, Mr. X, a HNI, send a cheque of ₹ 50,00,000 to the Fund and Fund Manager immediately purchases 18000 shares of C Ltd. and balance is held in bank. Then what will be position of fund.

(c) Now suppose on 2 April 2009 at 4.00 p.m. the market price of shares is as follows:

Shares	₹
A Ltd.	20.30
B Ltd.	513.70
C Ltd.	290.80
D Ltd.	671.90
E Ltd.	44.20

Then what will be new NAV.

Solution — ICAI

9. (a) NAV of the Fund.

$$= \frac{₹\,1,97,000 + ₹\,2,41,30,000 + ₹\,26,44,000 + ₹\,6,74,90,000 + ₹\,7,77,000}{800000}$$

$$= \frac{₹\,9,52,38,000}{800000} = ₹\ 119.0475 \text{ rounded to } ₹\ 119.05$$

(b) The revised position of fund shall be as follows:

Shares	No. of shares	Price	Amount (₹)
A Ltd.	10000	19.70	1,97,000
B Ltd.	50000	482.60	2,41,30,000
C Ltd.	28000	264.40	74,03,200
D Ltd.	100000	674.90	674,90,000
E Ltd.	30000	25.90	7,77,000
Cash			2,40,800
			10,02,38,000

$$\text{No. of units of fund} = 800000 + \frac{5000000}{119.0475} = 842000$$

(c) On 2nd April 2009, the NAV of fund will be as follows:

Shares	No. of shares	Price	Amount (₹)
A Ltd.	10000	20.30	2,03,000
B Ltd.	50000	513.70	2,56,85,000
C Ltd.	28000	290.80	81,42,400
D Ltd.	100000	671.90	6,71,90,000
E Ltd.	30000	44.20	13,26,000
Cash			2,40,800
			10,27,87,200

$$\text{NAV as on 2nd April 2009} = \frac{₹\,10,27,87,200}{842000} = ₹\ 122.075 \text{ per unit}$$

Practical Q10

10. Mr. Sinha has invested in three Mutual fund schemes as per details below:

	Scheme X	Scheme Y	Scheme Z
Date of Investment	01.12.2008	01.01.2009	01.03.2009
Amount of Investment	₹ 5,00,000	₹ 1,00,000	₹ 50,000
Net Asset Value at entry date	₹ 10.50	₹ 10.00	₹ 10.00
Dividend received upto 31.03.2009	₹ 9,500	₹ 1,500	Nil
NAV as at 31.3.2009	₹ 10.40	₹ 10.10	₹ 9.80

You are required to calculate the effective yield on per annum basis in respect of each of the three schemes to Mr. Sinha upto 31.03.2009.

Solution — ICAI

10. Calculation of effective yield on per annum basis in respect of three mutual fund schemes to Mr. Sinha up to 31-03-2009:

Particulars		MF X	MF Y	MF Z
(a)	Investments	₹ 5,00,000	₹ 1,00,000	₹ 50,000
(b)	Opening NAV	₹10.50	₹10.00	₹10.00
(c)	No. of units (a/b)	47,619.05	10,000	5,000
(d)	Unit NAV ON 31-3-2009	₹ 10.40	₹ 10.10	₹ 9.80
(e)	Total NAV on 31-3-2009 (c x d)	₹ 4,95,238.12	₹ 1,01,000	₹ 49,000
(f)	Increase / Decrease of NAV (e - a)	(₹ 4,761.88)	₹ 1,000	(₹ 1,000)
(g)	Dividend Received	₹ 9,500	₹ 1,500	Nil
(h)	Total yield (f + g)	₹ 4,738.12	₹ 2,500	(₹ 1,000)
(i)	Number of Days	121	90	31
(j)	Effective yield p.a. (h/a x 365/i x 100)	2.859%	10.139%	(-) 23.55%

Practical QII

11. Mr. Y has invested in the three mutual funds (MF) as per the following details:

Particulars	MF 'X'	MF 'Y'	MF 'Z'
Amount of Investment (₹)	2,00,000	4,00,000	2,00,000
Net Assets Value (NAV) at the time of purchase (₹)	10.30	10.10	10
Dividend Received up to 31.03.2018 (₹)	6,000	0	5,000
NAV as on 31.03.2018 (₹)	10.25	10	10.20
Effective Yield per annum as on 31.03.2018 (percent)	9.66	-11.66	24.15

Assume 1 Year =365 days

Mr. Y has misplaced the documents of his investment. Help him in finding the date of his original investment after ascertaining the following:

(i) Number of units in each scheme;

(ii) Total NAV;

(iii) Total Yield; and

(iv) Number of days investment held.

Solution — ICAI

11. (i) Number of Units in each Scheme

MF 'X'	₹ 2,00,000 / ₹ 10.30	= 19,417.48
MF 'Y'	₹ 4,00,000 / ₹ 10.10	= 39,603.96
MF 'Z'	₹ 2,00,000 / ₹10.00	= 20,000.00

(ii) Total NAV on 31.03.2018

MF 'X'	= 19,417.48 x ₹ 10.25	₹ 1,99,029.17
MF 'Y'	= 39,603.96 x ₹ 10.00	₹ 3,96,039.60
MF 'Z'	= 20,000.00 x ₹ 10.20	₹ 2,04,000.00
Total		₹ 7,99,068.77

(iii) Total Yield

	Capital Yield	Dividend Yield	Total
MF 'X'	₹ 1,99,029.17- ₹ 2,00,000 = - ₹ 970.83	₹ 6,000	₹ 5,029.17
MF 'Y'	₹ 3,96,039.60 - ₹ 4,00,000 = - ₹ 3,960.40	Nil	- ₹ 3,960.40
MF 'Z'	₹ 2,04,000 - ₹ 2,00,000 = ₹ 4,000	₹ 5,000	₹ 9,000.00
Total			₹ 10,068.77

$$\text{Total Yield} = \frac{₹10,068.77}{₹\ 8,00,000} \times 100 = 1.2586\%$$

(iv) No. of Days Investment Held

	MF 'X'	MF 'Y'	MF 'Z'
Initial Investment (₹)	2,00,000	4,00,000	2,00,000
Yield (₹)	5,029.17	-3,960.40	9,000.00
Yield (%)	2.5146	- 0.9901	4.5
Period of Holding (Days)	$\frac{2.5146}{9.66} \times 365$ = 95 Days	$\frac{-0.9901}{-11.66} \times 365$ = 31 Days	$\frac{4.5}{24.15} \times 365$ = 68 Days
Date of Original Investment	27.12.17	01.03.18	23.01.18

Alternatively following dates can also be considered:

Date of Original Investment	26.12.17	28.02.18	22.01.18

Practical Q12

12. Mr. X on 1.7.2007, during the initial offer of some Mutual Fund invested in 10,000 units having face value of ₹ 10 for each unit. On 31.3.2008, the dividend paid by the M.F. was 10% and Mr. X found that his annualized yield was 153.33%. On 31.3.2009, 20% dividend was given.

On 31.3.2010, Mr. X redeemed all his balance of 11,296.11 units when his annualized yield was 73.52%. What are the NAVs as on 31.3.2008, 31.3.2009 and 31.3.2010?

Solution — ICAI

12. Yield for 9 months = (153.33 x 9/12) = 115%

Market value of Investments as on 31.03.2008 = 1,00,000/- + (1,00,000x 115%)

= ₹2,15,000/-

Therefore, NAV as on 31.03.2008 = (2,15,000-10,000)/10,000= ₹20.50

(NAV would stand reduced to the extent of dividend payout, being (10,000x10x10%) = ₹10,000)

Since dividend was reinvested by Mr. X, additional units acquired = $\frac{₹10,000}{₹\ 20.50}$ = 487.80 units

Therefore, units as on 31.03.2008 = 10, 000+ 487.80 = 10,487.80

Alternately, units as on 31.03.2008 = (2,15,000/20.50) = 10,487.80]

Dividend as on 31.03.2009 = 10,487.80 x 10 x 0.2 = ₹20,975.60

Let X be the NAV on 31.03.2009, then number of new units reinvested will be ₹ 20,975.60/X. Accordingly 11296.11 units shall consist of reinvested units and 10487.80 (as on 31.03.2008). Thus, by way of equation it can be shown as follows:

$$11296.11 = \frac{20975.60}{X} + 10487.80$$

Therefore, NAV as on 31.03.2009 = 20,975.60/(11,296.11- 10,487.80)

= ₹25.95

NAV as on 31.03.2010 = ₹ 1,00,000 (1+0.7352x33/12)/11296.11

= ₹ 26.75

Practical Q13

13. Mr. X on 1.7.2012, during the initial public offer of a Mutual Fund (MF) invested ₹ 1,00,000 at Face Value of ₹ 10. On 31.3.2013, the MF declared a dividend of 10% when Mr. X calculated that his holding period return was 115%. On 31.3.2014, MF again declared a dividend of 20%. On 31.3.2015, Mr. X redeemed all his investment which had accumulated to 11,296.11 units when his holding period return was 202.17%.

Calculate the NAVs as on 31.03.2013, 31.03.2014 and 31.03.2015.

Solution — ICAI

13. Yield for 9 months = 115%

Market value of Investments as on 31.03.2013 = 1,00,000 + (1,00,000 x 115%)

= ₹ 2,15,000/-

Therefore, NAV as on 31.03.2013 = (2,15,000 -10,000)/10,000 = ₹ 20.50

(NAV would stand reduced to the extent of dividend payout, being (₹100,000 x 10%)

= ₹ 10,000)

Since dividend was reinvested by Mr. X, additional units acquired

$$= \frac{₹\ 10,000}{₹\ 20.50} = 487.80 \text{ units}$$

Therefore, units as on 31.03.2013 = 10,000+ 487.80 = 10,487.80

[Alternately, units as on 31.03.2013 = (2,15,000/20.50) = 10,487.80]

Dividend as on 31.03.2014 = 10,487.80 x 10 x 0.2 = ₹ 20,975.60

Let X be the NAV on 31.03.2014, then number of new units reinvested will be ₹ 20,975.60/X. Accordingly 11296.11 units shall consist of reinvested units and 10487.80 (as on 31.03.2013). Thus, by way of equation it can be shown as follows:

$$11296.11 = \frac{20975.60}{X} + 10487.80$$

Therefore, NAV as on 31.03.2014 = 20,975.60/(11,296.11- 10,487.80)

= ₹ 25.95

NAV as on 31.03.2015 = ₹ 1,00,000 (1+2.0217)/11296.11

= ₹ 26.75

Practical Q14

14. A Mutual Fund having 300 units has shown its NAV of ₹ 8.75 and ₹ 9.45 at the beginning and at the end of the year respectively. The Mutual Fund has given two options:

(i) Pay ₹ 0.75 per unit as dividend and ₹ 0.60 per unit as a capital gain, or

(ii) These distributions are to be reinvested at an average NAV of ₹ 8.65 per unit.

What difference would it make in terms of return available and which option is preferable?

Solution — ICAI

14. **(i)** Returns for the year

(All changes on a Per -Unit Basis)

Change in Price:	₹ 9.45 – ₹8.75 = ₹ 0.70
Dividends received:	₹ 0.75
Capital gains distribution	₹ 0.60
Total reward	₹ 2.05
Holding period reward:	$\frac{₹2.05}{₹8.75} \times 100 = 23.43\%$

(ii) When all dividends and capital gains distributions are re-invested into additional units of the fund @ (₹ 8.65/unit)

= Dividend + Capital Gains per unit

= ₹ 0.75 + ₹ 0.60 = ₹ 1.35

Total received from 300 units = ₹1.35 x 300 = ₹405/-.

Additional Units Acquired

= ₹405/₹8.65 = 46.82 Units.

Total No.of Units_ = 300 units + 46.82 units = 346.82 units.

Value of 346.82 units held at the end of the year

= 346.82 units x ₹9.45 = ₹3277.45

Price Paid for 300 Units at the beginning of the year

= 300 units x ₹8.75 = ₹2,625.00

Holding Period Reward

₹ (3277.45 – 2625.00) = ₹652.45

Holding Period Reward $= \frac{₹\ 652.45}{₹\ 2625.00} \times 100 = 24.85\%$

Conclusion: Since the holding period reward is more in terms of percentage in option-two i.e., reinvestment of distributions at an average NAV of ₹8.65 per unit, this option is preferable.

Practical QIS

15. On 1-4-2012 ABC Mutual Fund issued 20 lakh units at ₹ 10 per unit. Relevant initial expenses involved were ₹ 12 lakhs. It invested the funds so raised in capital market instruments to build a portfolio of ₹ 185 lakhs. During the month of April 2012 it disposed off some of the instruments costing ₹ 60 lakhs for ₹ 63 lakhs and used the proceeds in purchasing securities for ₹ 56 lakhs. Fund management expenses for the month of April 2012 was ₹ 8 lakhs of which 10% was in arrears. In April 2012 the fund earned dividends amounting to ₹ 2 lakhs and it distributed 80% of the realized earnings. On 30-4-2012 the market value of the portfolio was ₹ 198 lakhs.

Mr. Akash, an investor, subscribed to 100 units on 1-4-2012 and disposed off the same at closing NAV on 30-4-2012. What was his annual rate of earning?

Solution — ICAI

15.

	Amount in ₹ lakhs	Amount in ₹ lakhs	Amount in ₹ lakhs
Opening Bank (200 - 185 -12)	3.00		
Add: Proceeds from sale of securities	63.00		
Add: Dividend received	2.00	68.00	
Deduct:			
Cost of securities purchased	56.00		
Fund management expenses paid (90% of 8)	7.20		
Capital gains distributed = 80% of (63 – 60)	2.40		
Dividend distributed =80% of 2.00	1.60	67.20	
Closing Bank			0.80
Closing market value of portfolio			198.00
			198.80
Less: Arrears of expenses			0.80
Closing Net Assets			198.00
Number of units (Lakhs)			20
Closing NAV per unit (198.00/20)			9.90

Rate of Earning (Per Unit)

	Amount
Income received (₹ 2.40 + ₹ 1.60)/20	₹ 0.20
Loss: Loss on disposal (₹ 200 - ₹ 198)/20	₹ 0.10
Net earning	₹ 0.10
Initial investment	₹ 10.00
Rate of earning (monthly)	1%
Rate of earning (Annual)	12%

Practical Q16

16. Sun Moon Mutual Fund (Approved Mutual Fund) sponsored open-ended equity-oriented scheme "Chanakya Opportunity Fund". There were three plans viz. 'A' – Dividend Re-investment Plan, 'B' – Bonus Plan & 'C' – Growth Plan.

At the time of Initial Public Offer on 1.4.1999, Mr. Anand, Mr. Bacchan & Mrs. Charu, three investors invested ₹ 1,00,000 each & chose 'B', 'C' & 'A' Plan respectively.

The History of the Fund is as follows:

Date	Dividend %	Bonus Ratio	Net Asset Value per Unit (F.V. ₹ 10)		
			Plan A	Plan B	Plan C

28.07.2003	20		30.70	31.40	33.42
31.03.2004	70	5 : 4	58.42	31.05	70.05
31.10.2007	40		42.18	25.02	56.15
15.03.2008	25		46.45	29.10	64.28
31.03.2008		1 : 3	42.18	20.05	60.12
24.03.2009	40	1 : 4	48.10	19.95	72.40
31.07.2009			53.75	22.98	82.07

On 31st July all three investors redeemed all the balance units.

Calculate annual rate of return to each of the investors.

Consider:

1. Long-term Capital Gain is exempt from Income tax.
2. Short-term Capital Gain is subject to 10% Income tax.
3. Security Transaction Tax 0.2 per cent only on sale/redemption of units.
4. Ignore Education Cess.

Solution — ICAI

16. **Mrs. Charu Plan A Dividend Reinvestment** *(Amount in ₹)*

Date	Investment (₹)	Dividend payout (%)	Dividend Re-invested (Closing Units X Face value of '10 X Dividend Payout %)	NAV (₹)	Units	Closing Unit Balance ∑ Units
01.04.1999	1,00,000.00			10.00	10,000.00	10,000.00
28.07.2003		20	20,000.00	30.70	651.47	10,651.47
31.03.2004		70	74,560.29	58.42	1,276.28	11,927.75
30.10.2007		40	47,711.00	42.18	1,131.13	13,058.88
15.03.2008		25	32,647.20	46.45	702.85	13,761.73
24.03.2009		40	55,046.92	48.10	1,144.43	14,906.16
Redemption value 14,906.16 × 53.75						8,01,206.10
Less: Security Transaction Tax (STT) is 0.2%						1,602.41
Net amount received						7,99,603.69
Less: Short term capital gain tax @ 10% on 1,144.43 (53.64* – 48.10≈) = 6,340						634
Net of tax						7,98,969.69
Less: Investment						1,00,000.00
						6,98,969.69

*(53.75 – STT @ 0.2%)

≈ This value can also be taken as zero

$$\text{Annual average return (\%)} = \frac{6,98,969.69}{1,00,000} \times \frac{12}{124} \times 100 = 67.64\ \%$$

Mr. Anand Plan B – Bonus

(Amount in ₹)

Date	Units	Bonus units	Total Balance	NAV per unit
01.04.1999	10,000		10,000	10
31.03.2004		12,500	22,500	31.05
31.03.2008		7,500	30,000	20.05
24.03.2009		7,500	37,500	19.95

Redemption value 37,500 × 22.98	8,61,750.00
Less: Security Transaction Tax (STT) is 0.2%	1,723.50
Net amount received	8,60,026.50
Less: Short term capital gain tax @ 10%	
7,500 × (22.93† – 19.95≈) = 22,350	2,235.00
Net of tax	8,57,791.50
Less: Investment	1,00,000.00
Net gain	7,57,791.50

†(22.98 – STT @ 0.2%)

≈ This value can also be taken as zero

$$\text{Annual average return (\%)} = \frac{7,57,791.50}{1,00,000} \times \frac{12}{124} \times 100 = 73.33\ \%$$

Mr. Bacchan Plan C – Growth

Particulars	*(Amount in ₹)*
Redemption value 10,000 × 82.07	8,20,700.00
Less: Security Transaction Tax (S.T.T) is .2%	1,641.40
Net amount received	8,19,058.60
Less: Short term capital gain tax @ 10%	0.00
Net of tax	8,19,058.60
Less: Investment	1,00,000.00
Net gain	7,19,058.60

$$\text{Annual average return (\%)} = \frac{7,19,058}{1,00,000} \times \frac{12}{124} \times 100 = 69.59\%$$

Note: Alternatively, figure of * and † can be taken as without net of Tax because, as per Proviso 5 of Section 48 of IT Act, no deduction of STT shall be allowed in computation of Capital Gain.

Practical Q17

17. A mutual fund company introduces two schemes i.e. Dividend plan (Plan-D) and Bonus plan (Plan-B). The face value of the unit is ₹ 10. On 1-4-2005 Mr. K invested ₹ 2,00,000 each in Plan-D and Plan-B when the NAV was ₹ 38.20 and ₹ 35.60 respectively. Both the plans matured on 31-3-2010.

Particulars of dividend and bonus declared over the period are as follows:

Date	Dividend	Bonus Ratio	Net Asset Value (₹)	
	%		Plan D	Plan B
30-09-2005	10	---	39.10	35.60
30-06-2006	---	1:5	41.15	36.25
31-03-2007	15	---	44.20	33.10
15-09-2008	13	---	45.05	37.25
30-10-2008	---	1:8	42.70	38.30
27-03-2009	16	---	44.80	39.10
11-04-2009	---	1:10	40.25	38.90
31-03-2010	---	---	40.40	39.70

What is the effective yield per annum in respect of the above two plans?

Solution — ICAI

17. **Plan – D**

Unit acquired = $\frac{2,00,000}{38.20}$ = 5235.60

Date	Units held	Dividend		Reinvestment	New	Total
		%	Amount	Rate	Units	Units
01.04.2005						5235.60
30.09.2005	5235.60	10	5235.60	39.10	133.90	5369.50
31.03.2007	5369.50	15	8054.25	44.20	182.22	5551.72
15.09.2008	5551.72	13	7217.24	45.05	160.20	5711.92
27.03.2009	5711.92	16	9139.07	44.80	204	5915.92
31.03.2010	Maturity Value		(₹ 40.40 X 5915.92)			₹ 2,39,003.17
	Less: Cost of Acquisition					₹ 2,00,000.00
	Total Gain					₹ 39,003.17

$$\therefore \text{Effective Yield} = \frac{₹39,003.17}{₹2,00,000} \times \frac{1}{5} \times 100 = 3.90\%$$

Alternatively, it can be computed by using the IRR method as follows:

NPV at 4% = - ₹ 2,00,000 + ₹ 1,96,443 = - ₹ 3,557

NPV at 2% = - ₹ 2,00,000 + ₹ 2,16,473 = ₹ 16,473

$$\text{IRR} = \text{LR} + \frac{\text{NPV at LR}}{\text{NPV at LR - NPV at HR}} (\text{HR - LR}) = 2\% + \frac{16473}{16473 - (-3557)} (4\% - 2\%) = 3.645\%$$

Plan – B

Date	Particulars	Calculation Working	No. of Units	NAV (₹)
1.4.05	Investment	₹2,00,000/35.60=	5617.98	35.60
30.6.06	Bonus	5617.98/5 =	1123.60	36.25
			6741.58	
30.10.08	"	6741.58/8 =	842.70	38.30
			7584.28	
11.4.09	"	7584.28/10 =	758.43	38.90
			8342.71	
31.3.10	Maturity Value	8342.71 x ₹ 39.70=		3,31,205.59
	Less: Investment			2,00,000.00
	Gain			1,31,205.59

$$\therefore \text{Effective Yield} = \frac{1,31,205.59}{2,00,000} \times \frac{1}{5} \times 100 = 13.12\%$$

Alternatively, it can be computed by using the IRR method as follows:

NPV at 13% = - ₹ 2,00,000 + ₹ 1,79,765 = - ₹ 20,235

NPV at 8% = - ₹ 2,00,000 + ₹ 2,25,413 = ₹ 25,413

$$\text{IRR} = \text{LR} + \frac{\text{NPV at LR}}{\text{NPV at LR - NPV at HR}} (\text{HR - LR}) = 8\% + \frac{25413}{25413 - (-20235)} (13\% - 8\%) = 10.78\%$$

Practical Q18

18. A mutual fund made an issue of 10,00,000 units of ₹ 10 each on January 01, 2008. No entry load was charged. It made the following investments:

Particulars	₹
50,000 Equity shares of ₹ 100 each @ ₹ 160	80,00,000
7% Government Securities	8,00,000
9% Debentures (Unlisted)	5,00,000
10% Debentures (Listed)	5,00,000
	98,00,000

During the year, dividends of ₹ 12,00,000 were received on equity shares. Interest on all types of debt securities was received as and when due. At the end of the year equity shares and 10% debentures are quoted at 175% and 90% respectively of face value. Other investments are valued at par.

Find out the Net Asset Value (NAV) per unit given that operating expenses paid during the year amounted to ₹ 5,00,000. Also find out the NAV, if the Mutual fund had distributed a dividend of ₹ 0.80 per unit during the year to the unit holders.

Solution — ICAI

18. In order to find out the NAV, the cash balance at the end of the year is calculated as follows-

Particulars	₹
Cash balance in the beginning	
(₹ 100 lakhs – ₹ 98 lakhs)	2,00,000
Dividend Received	12,00,000
Interest on 7% Govt. Securities	56,000
Interest on 9% Debentures	45,000
Interest on 10% Debentures	50,000
	15,51,000
(-) Operating expenses	5,00,000
Net cash balance at the end	10,51,000
Calculation of NAV	₹
Cash Balance	10,51,000
7% Govt. Securities (at par)	8,00,000
50,000 equity shares @ ₹ 175 each	87,50,000
9% Debentures (Unlisted) at cost	5,00,000
10% Debentures @90%	4,50,000
Total Assets	1,15,51000
No. of Units	10,00,000
NAV per Unit	₹ 11.55

Calculation of NAV, if dividend of ₹ 0.80 is paid –

Net Assets (₹ 1,15,51,000 – ₹ 8,00,000)	₹ 1,07,51,000
No. of Units	10,00,000
NAV per unit	₹ 10.75

Practical Q19

19. Based on the following information, determine the NAV of a regular income scheme on per unit basis:

Particulars	₹ Crores
Listed shares at Cost (ex-dividend)	20
Cash in hand	1.23
Bonds and debentures at cost	4.3
Of these, bonds not listed and quoted	1
Other fixed interest securities at cost	4.5
Dividend accrued	0.8
Amount payable on shares	6.32
Expenditure accrued	0.75
Number of units (₹ 10 face value)	20 lacs
Current realizable value of fixed income securities of face value of ₹ 100	106.50
The listed shares were purchased when Index was	1,000
Present index is	2,300
Value of listed bonds and debentures at NAV date	8

There has been a diminution of 20% in unlisted bonds and debentures. Other fixed interest securities are at cost.

Solution — ICAI

19.

Particulars	Adjusted Values ₹ crores
Equity Shares	46.00
Cash in hand	1.23
Bonds and debentures not listed	0.80
Bonds and debentures listed	8.00
Dividends accrued	0.80
Fixed income securities	4.50
Sub total assets (A)	**61.33**
***Less:* Liabilities**	
Amount payable on shares	6.32
Expenditure accrued	0.75
Sub total liabilities (B)	**7.07**
Net Assets Value (A) – (B)	**54.26**
No. of units	20,00,000
Net Assets Value per unit (₹ 54.26 crore / 20,00,000)	₹ 271.30

Practical Q20

20. On 1st April, an open ended scheme of mutual fund had 300 lakh units outstanding with Net Assets Value (NAV) of ₹ 18.75. At the end of April, it issued 6 lakh units at opening NAV plus 2% load, adjusted for dividend equalization. At the end of May, 3 Lakh units were repurchased at opening NAV less 2% exit load adjusted for dividend equalization. At the end of June, 70% of its available income was distributed.

In respect of April-June quarter, the following additional information are available:

	₹ in lakh
Portfolio value appreciation	425.47
Income of April	22.950
Income for May	34.425
Income for June	45.450

You are required to calculate:

(i) Income available for distribution;

(ii) Issue price at the end of April;

(iii) repurchase price at the end of May; and

(iv) net asset value (NAV) as on 30th June.

Solution — ICAI

20. Calculation of Income available for Distribution

	Units (Lakh)	Per Unit (₹)	Total (₹ In lakh)
Income from April	300	0.0765	22.9500
Add: Dividend equalization collected on issue	6	0.0765	0.4590
	306	0.0765	23.4090
Add: Income from May		0.1125	34.4250
	306	0.1890	57.8340
Less: Dividend equalization paid on repurchase	3	0.1890	(0.5670)
	303	0.1890	57.2670
Add: Income from June		0.1500	45.4500
	303	0.3390	102.7170
Less: Dividend Paid		0.2373	(71.9019)
	303	0.1017	30.8151

Calculation of Issue Price at the end of April

	₹
Opening NAV	18.750
Add: Entry Load 2% of ₹ 18.750	0.375
	19.125
Add: Dividend Equalization paid on Issue Price	0.0765
	19.2015

Calculation of Repurchase Price at the end of May

	₹
Opening NAV	18.750
Less: Exit Load 2% of ₹ 18.750	(0.375)
	18.375
Add: Dividend Equalization paid on Issue Price	0.1890
	18.564

Closing NAV

		₹ (Lakh)
Opening Net Asset Value (₹ 18.75 × 300)		5625.0000
Portfolio Value Appreciation		425.4700
Issue of Fresh Units (6 × 19.2015)		115.2090
Income Received (22.950 + 34.425 + 45.450)		102.8250
		6268.504
Less: Units repurchased (3 × 18.564)	-55.692	
Income Distributed	-71.9019	(-127.5939)
Closing Net Asset Value		6140.9101
Closing Units (300 + 6 – 3) lakh		303 lakh
∴ Closing NAV as on 30th June		₹ 20.2670

Practical Q21

21. There are two Mutual Funds viz. D Mutual Fund Ltd. and K Mutual Fund Ltd. Each having close ended equity schemes.

NAV as on 31-12-2014 of equity schemes of D Mutual Fund Ltd. is ₹ 70.71 (consisting 99% equity and remaining cash balance) and that of K Mutual Fund Ltd. is ₹ 62.50 (consisting 96% equity and balance in cash).

Following is the other information:

Particular	Equity Schemes	
	D Mutual Fund Ltd.	K Mutual Fund Ltd.
Sharpe Ratio	2	3.3
Treynor Ratio	15	15
Standard deviation	11.25	5

There is no change in portfolios during the next month and annual average cost is ₹ 3 per unit for the schemes of both the Mutual Funds.

If Share Market goes down by 5% within a month, calculate expected NAV after a month for the schemes of both the Mutual Funds.

For calculation, consider 12 months in a year and ignore number of days for particular month.

Solution — ICAI

21. Working Notes:

(i) Decomposition of Funds in Equity and Cash Components

	D Mutual Fund Ltd.	K Mutual Fund Ltd.
NAV on 31.12.14	₹ 70.71	₹ 62.50
% of Equity	99%	96%
Equity element in NAV	₹ 70	₹ 60
Cash element in NAV	₹ 0.71	₹ 2.50

(ii) Calculation of Beta

(a) D Mutual Fund Ltd.

$$\text{Sharpe Ratio} = 2 = \frac{E(R) - R_f}{\sigma_D} = \frac{E(R) - R_f}{11.25}$$

$$E(R) - R_f = 22.50$$

$$\text{Treynor Ratio} = 15 = \frac{E(R) - R_f}{\beta_D} = \frac{22.50}{\beta_D}$$

$$\beta_D = 22.50/15 = 1.50$$

(b) K Mutual Fund Ltd.

$$\text{Sharpe Ratio} = 3.3 = \frac{E(R) - R_f}{\sigma_K} = \frac{E(R) - R_f}{5}$$

$$E(R) - R_f = 16.50$$

$$\text{Treynor Ratio} = 15 = \frac{E(R) - R_f}{\beta_K} = \frac{16.50}{\beta_K}$$

$$\beta_K = 16.50/15 = 1.10$$

(iii) Decrease in the Value of Equity

	D Mutual Fund Ltd.	K Mutual Fund Ltd.
Market goes down by	5.00%	5.00%
Beta	1.50	1.10
Equity component goes down	7.50%	5.50%

(iv) Balance of Cash after 1 month

	D Mutual Fund Ltd.	K Mutual Fund Ltd.
Cash in Hand on 31.12.14	₹ 0.71	₹ 2.50
Less: Exp. Per month	₹ 0.25	₹ 0.25
Balance after 1 month	₹ 0.46	₹ 2.25

NAV after 1 month

	D Mutual Fund Ltd.	K Mutual Fund Ltd.
Value of Equity after 1 month		
70 x (1 - 0.075)	₹ 64.75	-
60 x (1 - 0.055)	-	₹ 56.70
Cash Balance	0.46	2.25
	65.21	58.95

Practical Q22

22. ANP Plan, a hedge fund currently has assets of ₹ 20 crore. CA. X, the manager of fund charges fee of 0.10% of portfolio asset. In addition to it he charges incentive fee of 2%. The incentive will be linked to gross return each year in excess of the portfolio maximum value since the inception of fund. The maximum value the fund achieved so far since inception of fund about one and half year ago was ₹ 21 crores.

You are required to compute the fee payable to CA. X, if return on the fund this year turns out to be

(a) 29%, (b) 4.5%, (c) -1.8%

Solution — ICAI

22. (a) If return is 29%

	₹
Fixed fee (A) 0.10% of ₹ 20 crore	2,00,000
New Fund Value (1.29 x ₹ 20 crore)	25.80 crore
Excess Value of best achieved (25.8 crore – 21.0 crore)	4.80 crore
Incentive Fee (2% of 4.80 crores) (B)	9,60,000
Total Fee (A)+(B)	11,60,000

(b) If return is 4.5%

	₹
Fixed (A) 0.10% of ₹ 20 crore	2,00,000
New Fund Value (1.045 x ₹ 20 crore)	20.90 crore
Excess Value of best achieved (20.90 crore –21.00 crore)	(₹ 0.10 crore)
Incentive Fee (as does not exceed best achieved) (B)	Nil
Total Fee (A)+(B)	2,00,000

(c) If return is (-1.8%)

No incentive only fixed fee of ₹ 2,00,000 will be paid

Practical Q23

23. Ms. Sunidhi is working with an MNC at Mumbai. She is well versant with the portfolio management techniques and wants to test one of the techniques on an equity fund she has constructed and compare the gains and losses from the technique with those from a passive buy and hold strategy. The fund consists of equities only and the ending NAVs of the fund he constructed for the last 10 months are given below:

Month Ending	NAV (₹/unit)	Month Ending	NAV (₹/unit)
December 2008	40.00	May 2009	37.00
January 2009	25.00	June 2009	42.00
February 2009	36.00	July 2009	43.00
March 2009	32.00	August 2009	50.00
April 2009	38.00	September 2009	52.00

Assume Sunidhi had invested a notional amount of ₹ 2 lakhs equally in the equity fund and a conservative portfolio (of bonds) in the beginning of December 2008 and the total portfolio was being rebalanced each time the NAV of the fund increased or decreased by 15%.

You are required to determine the value of the portfolio for each level of NAV following the Constant Ratio Plan.

Solution — ICAI

23. Constant Ratio Plan:

Stock Portfolio NAV (₹)	Value of Conservative Portfolio (₹)	Value of aggressive Portfolio (₹)	Total value of Constant Ratio Plan (₹)	Revaluation Action	Total No. of units in aggressive portfolio
40.00	1,00,000	1,00,000	2,00,000	-	2500
25.00	1,00,000	62,500	1,62,500	-	2500
	81,250	81,250	1,62,500	Buy 750 units	3250
36.00	81,250	1,17,000	1,98,250	-	3250
	99,125	99,125	1,98,250	Sell 496.53 units	2753.47
32.00	99,125	88,111.04	1,87,236.04	-	2753.47
38.00	99,125	1,04,631.86	2,03,756.86	-	2753.47
	1,01,878.43	1,01,878.43	2,03,756.86	Sell 72.46 units	2681.01
37.00	1,01,878.43	99,197.37	2,01,075.80	-	2681.01
42.00	1,01,878.43	1,12,602.42	2,14,480.85	-	2681.01
43.00	1,01,878.43	1,15,283.43	2,17,161.86	-	2681.01
50.00	1,01,878.43	1,34,050.50	2,35,928.93	-	2681.01
	1,17,964.47	1,17,964.46	2,35,928.93	Sell 321.72 units	2359.29
52.00	1,17,964.47	1,22,683.08	2,40,647.55	-	2359.29

Hence, the ending value of the mechanical strategy is ₹ 2,40,647.55 and buy & hold strategy is ₹ 2,60,000.

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